

Lab 3.3.1.3 Report: Monitor and Manage System Resources in Windows

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Objectives

The objectives of this lab were to:

- Learn how to start and stop Windows services
 - Monitor system resources using Performance Monitor
 - Use Computer Management utility to view system events
 - Create and configure Data Collector Sets for resource monitoring
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Part 1: Starting and Stopping the Routing and Remote Access ServiceSteps Performed:

1. Opened Network Connections window and left it visible
2. Launched Performance Monitor and cleared the default graph
3. Accessed Services console and located Routing and Remote Access service
4. Changed service startup type to Manual
5. Started the service while monitoring Performance Monitor
6. Observed changes in Network Connections
7. Stopped the service and observed the changes

Answers to Questions:

Question 1: What changes appear in the window after starting the Routing and Remote Access service?

Answer:

After refreshing the Network Connections window (F5), new virtual network adapters appeared including "Incoming Connections"

Question 2: What changes appear in the right pane after stopping the Routing and Remote Access service?

Answer:

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مثال: The virtual network adapters that appeared when the service started have disappeared or changed to a disabled/disconnected state. The Network Connections window returned to showing only the physical network adapters.

Question 3: Which Counter is being recorded the most in the graph?

Answer:

The counter being recorded the most is "% Processor Time" which appears in green color on the graph. This counter shows the percentage of CPU utilization over time.

Question 4: What values are displayed by the counter?

Answer:

The report view displays a numerical value representing the current percentage of processor time. At the time of observation, the value was approximately 12.456%, indicating low CPU usage.

Question 5: What is the Status and Startup Type for Routing and Remote Access?

Answer:

Status: Stopped

Startup Type: Disabled

Part 2: Working in the Computer Management Utility

Steps Performed:

1. Opened Computer Management from Administrative Tools
 2. Navigated to Event Viewer > Windows Logs > System
 3. Located events related to Routing and Remote Access service
 4. Analyzed the sequence of events for service start and stop operations
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Part 3: Configuring Administrative Tools

Steps Performed:

1. Created a new Data Collector Set named "Memory Logs"
2. Configured it to monitor "Available MBytes" counter
3. Set sampling interval to 4 seconds
4. Configured output location to C:\PerfLogs
5. Changed log format to Comma Separated (CSV)
6. Started data collection and forced memory usage by opening/closing browser
7. Stopped collection and analyzed the resulting CSV file

Answers to Questions:

Question 7: What is the full path name to the example file?

Answer:

C:\PerfLogs\OMARHAMED_20251013-000001\DataCollector01.csv

Question 8: What change do you notice for the Memory Logs icon?

Answer:

1. The green arrow overlay that was present on the Memory Logs icon during active collection disappeared, indicating that data collection is no longer active.
2. The icon returned to its normal state (without the green arrow)

. 3. Additionally, in the right pane output window, the path to the newly created log folder appeared

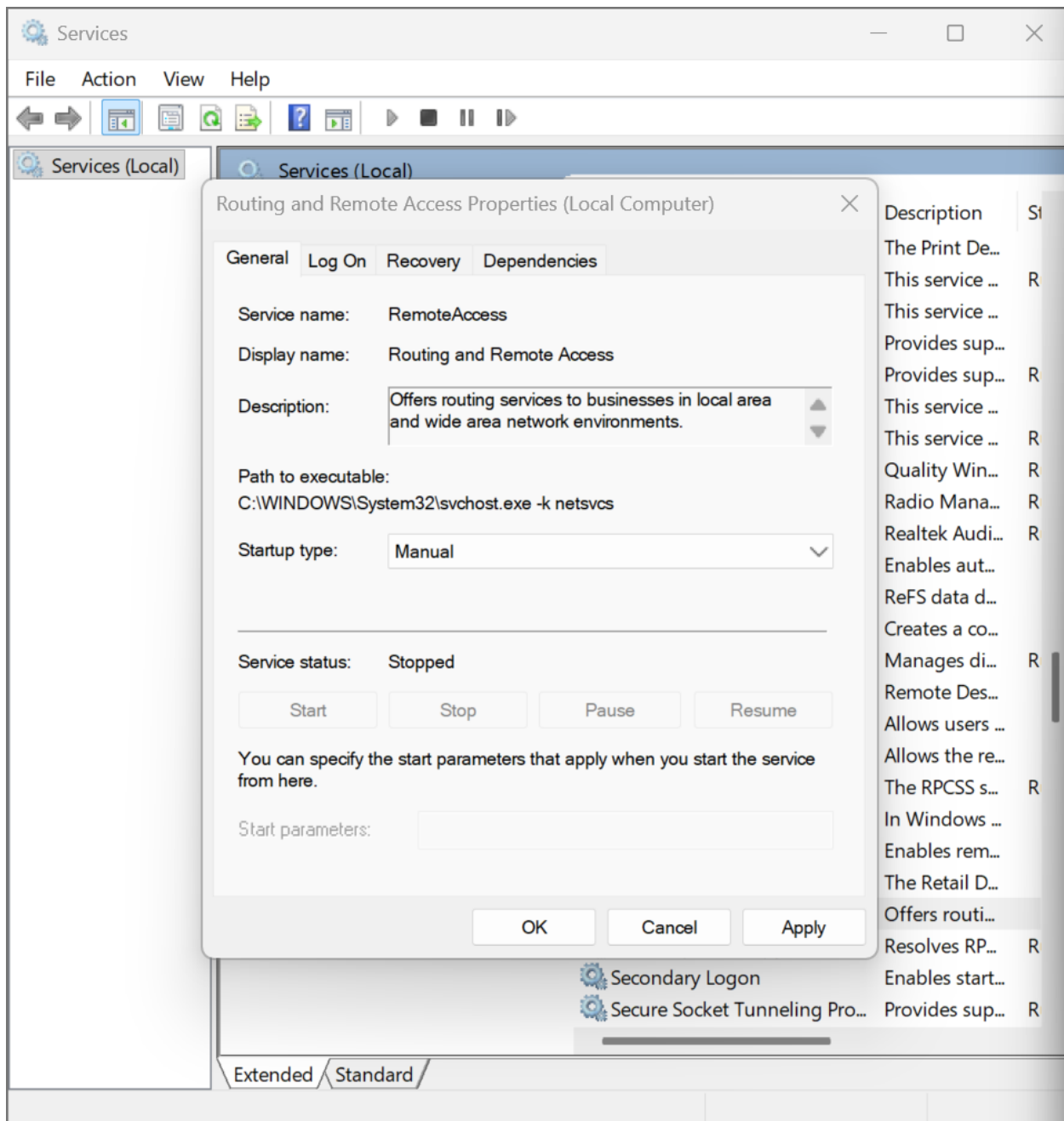
This confirms that the data collection was successfully completed and the CSV file was generated in the specified directory.

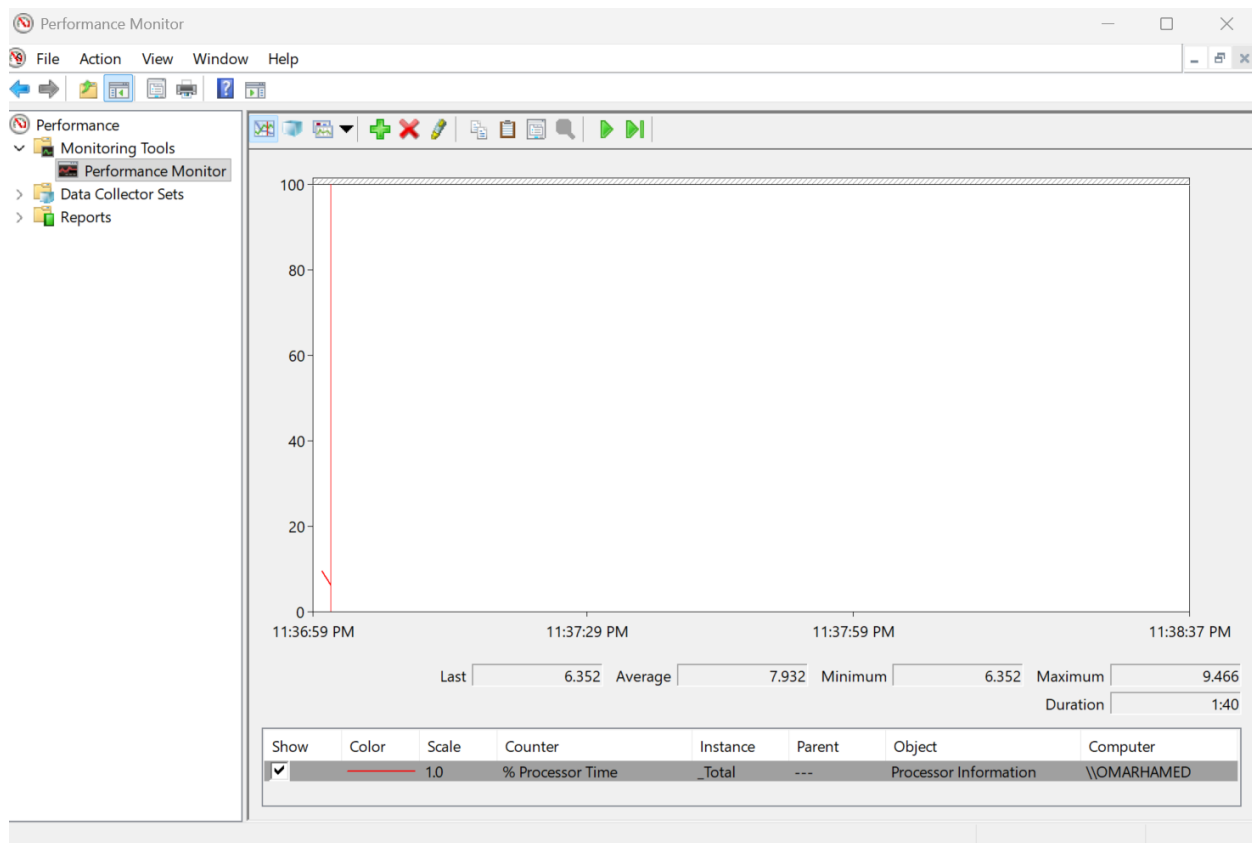
Question 9: What does the column farthest to the right show?

Answer:

The rightmost column displays the "Available MBytes" values, showing the amount of free RAM available in megabytes at each sampling interval (every 4 seconds).

Screenshots





Network Connections

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 << Netw... > Network Con...

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Search Network Connections 

Organize ▼

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Ethernet

Enabled

VirtualBox Host-Only Ethernet Ad...



Wi-Fi

omar home

RZ608 Wi-Fi 6E 80MHz



Incoming Connections

Local Area

ConnectionNetwork Configuratio...

No clients connected

3 items |







Network Connections



<< Netw... > Network Con...



Search Network Connections



Organize ▾



Ethernet

Enabled

VirtualBox Host-Only Ethernet Ad...



Wi-Fi

omar home

RZ608 Wi-Fi 6E 80MHz

2 items |





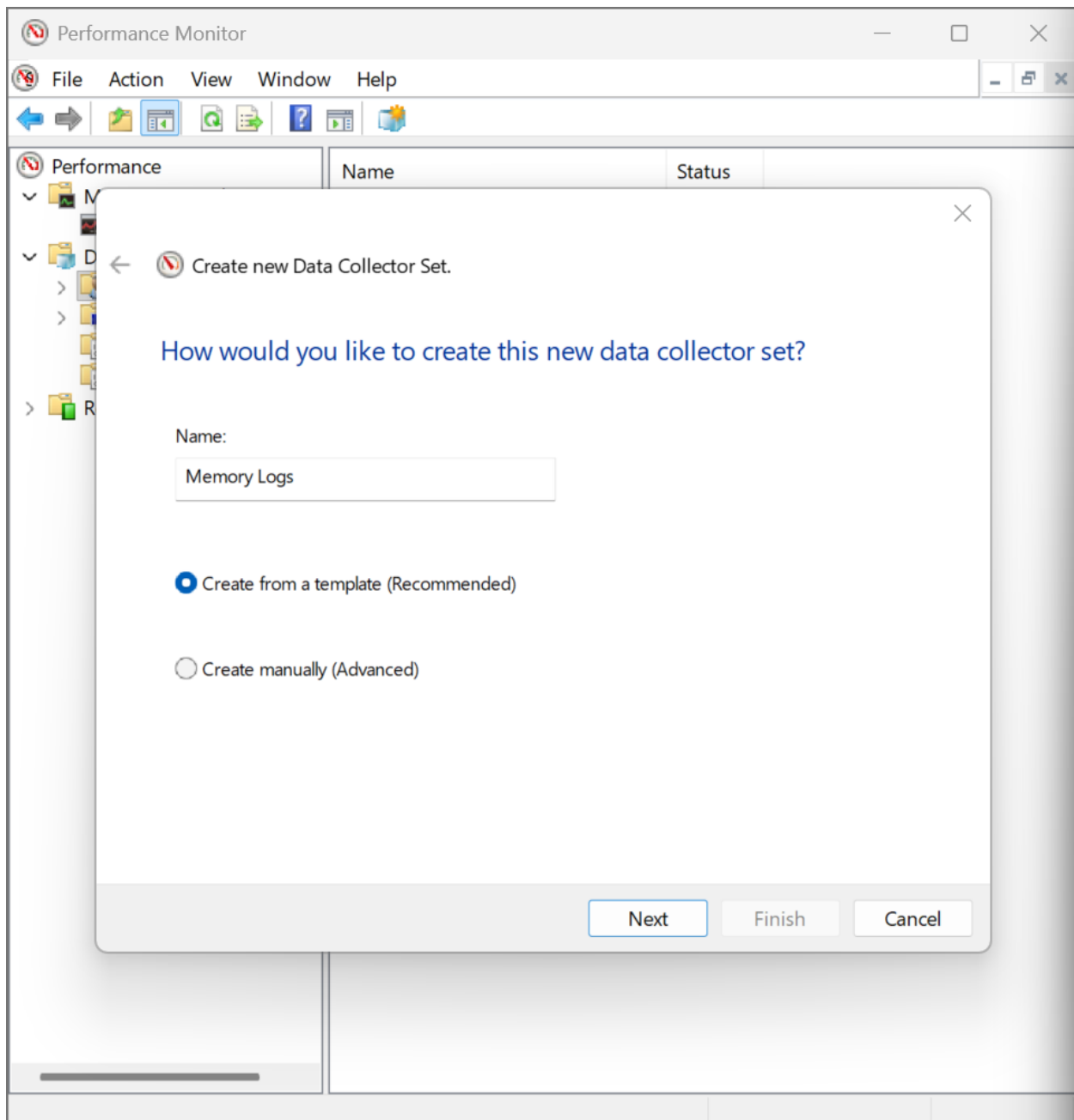
\\OMARHAMED

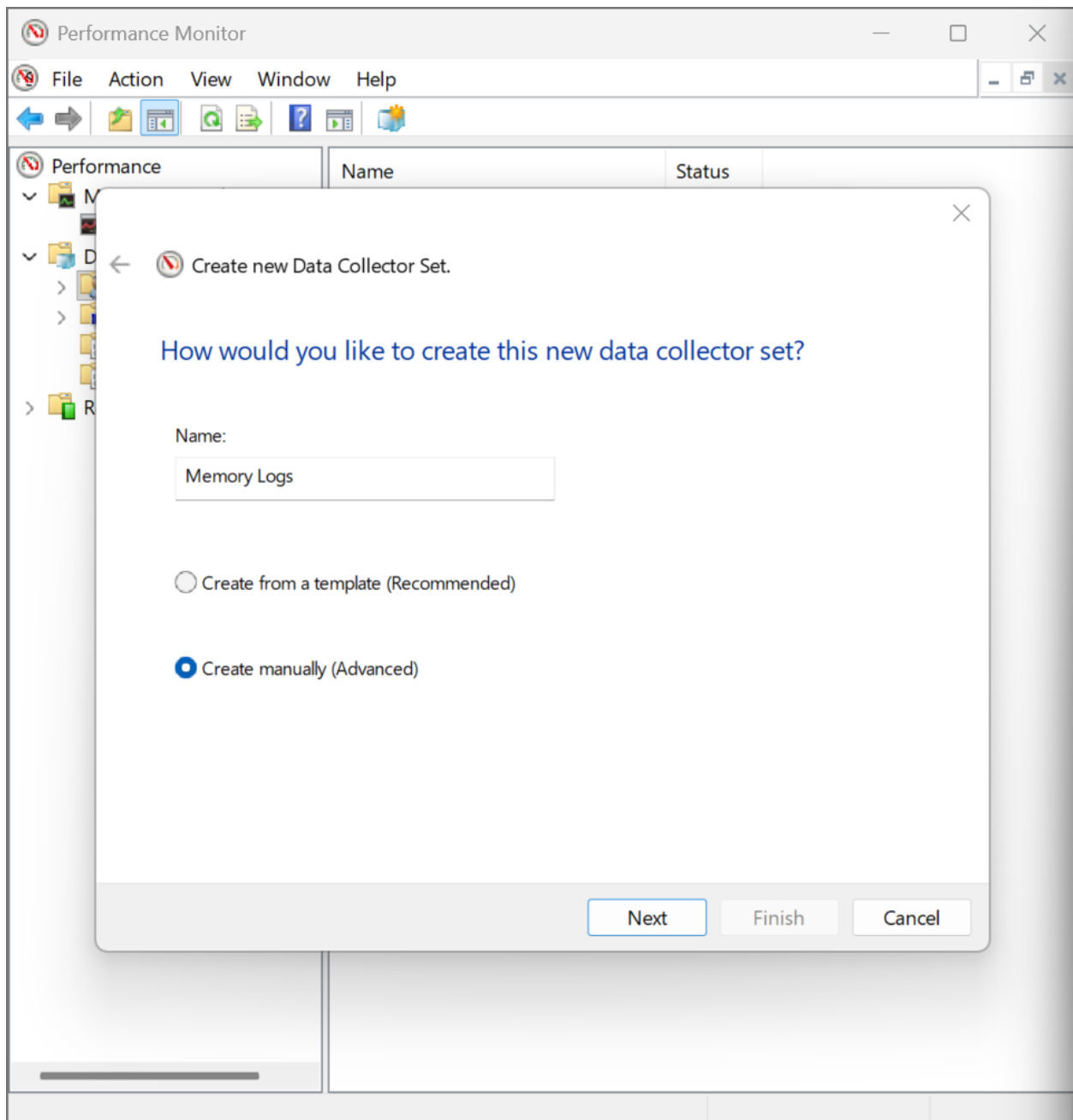
Processor Information

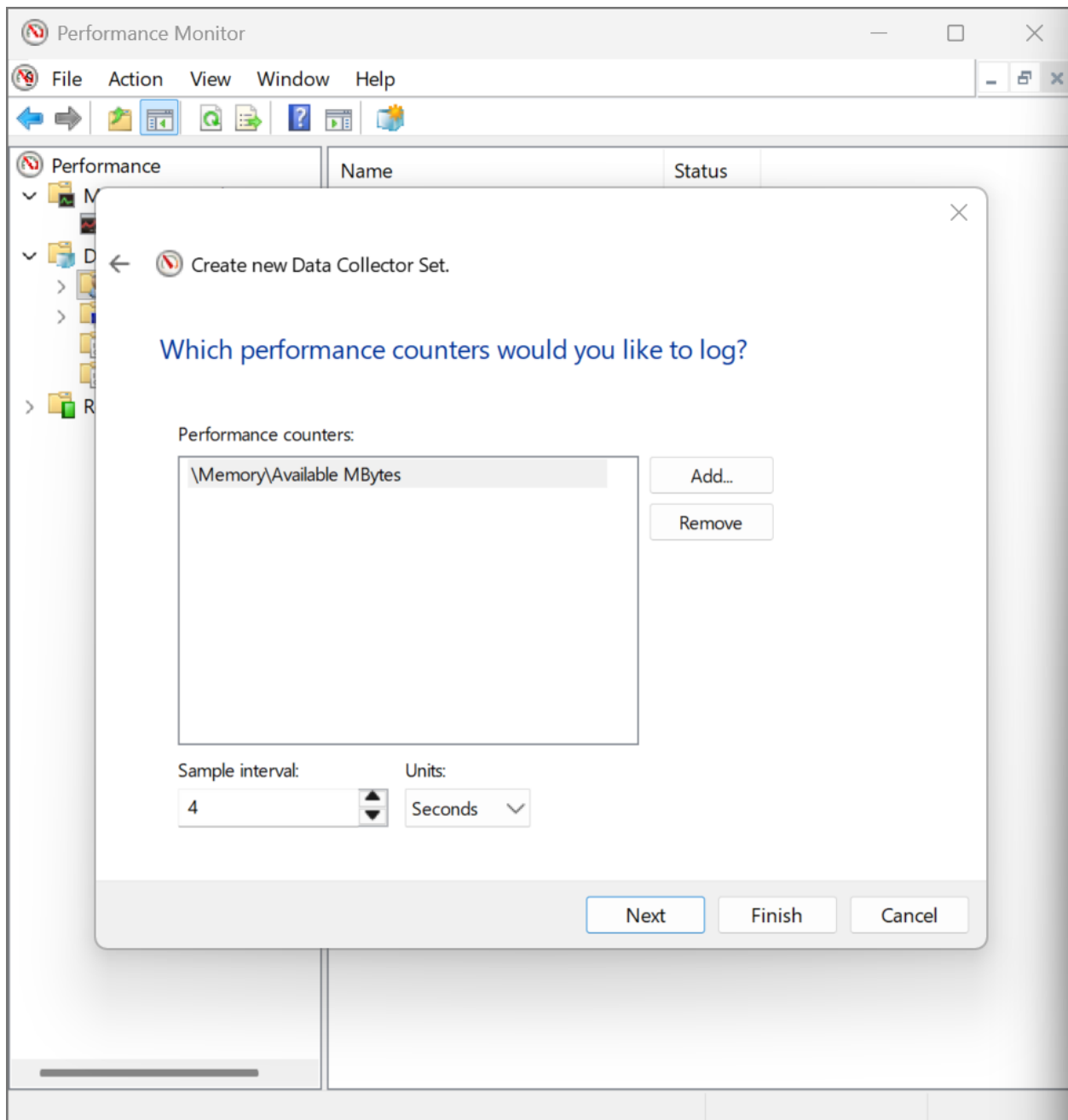
% Processor Time

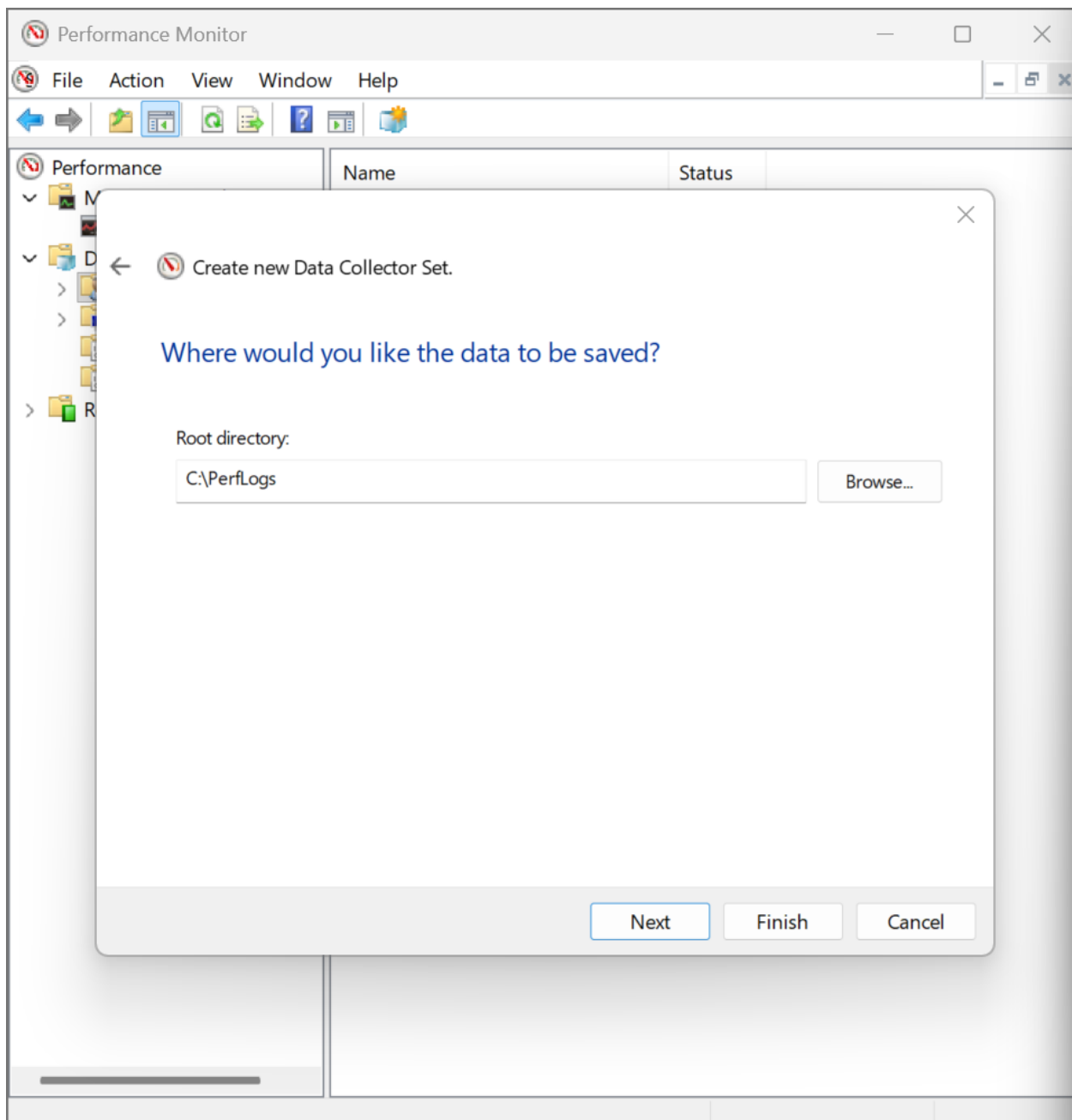
_Total

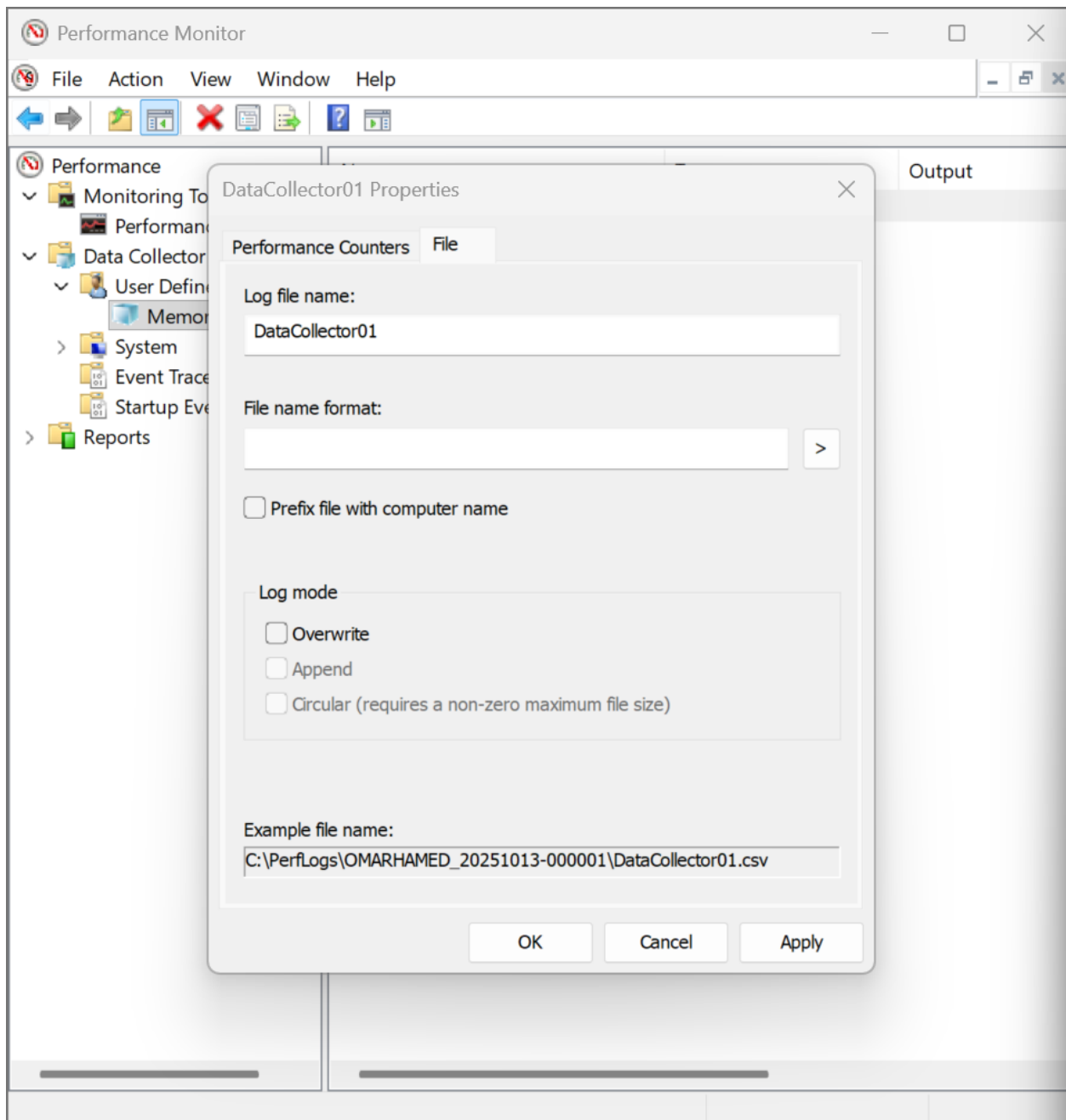
4.081











Conclusion

This lab successfully demonstrated the essential Windows system administration tools for monitoring and managing system resources. I learned how to:

- Control Windows services and understand their impact on the system
- Use Performance Monitor for real-time resource tracking

- Access Event Viewer to investigate system events and service activities
- Create custom Data Collector Sets for automated monitoring and analysis

These skills are crucial for system administrators to maintain system health, troubleshoot issues, and optimize performance. The hands-on experience with these tools provides a foundation for more advanced system management tasks.